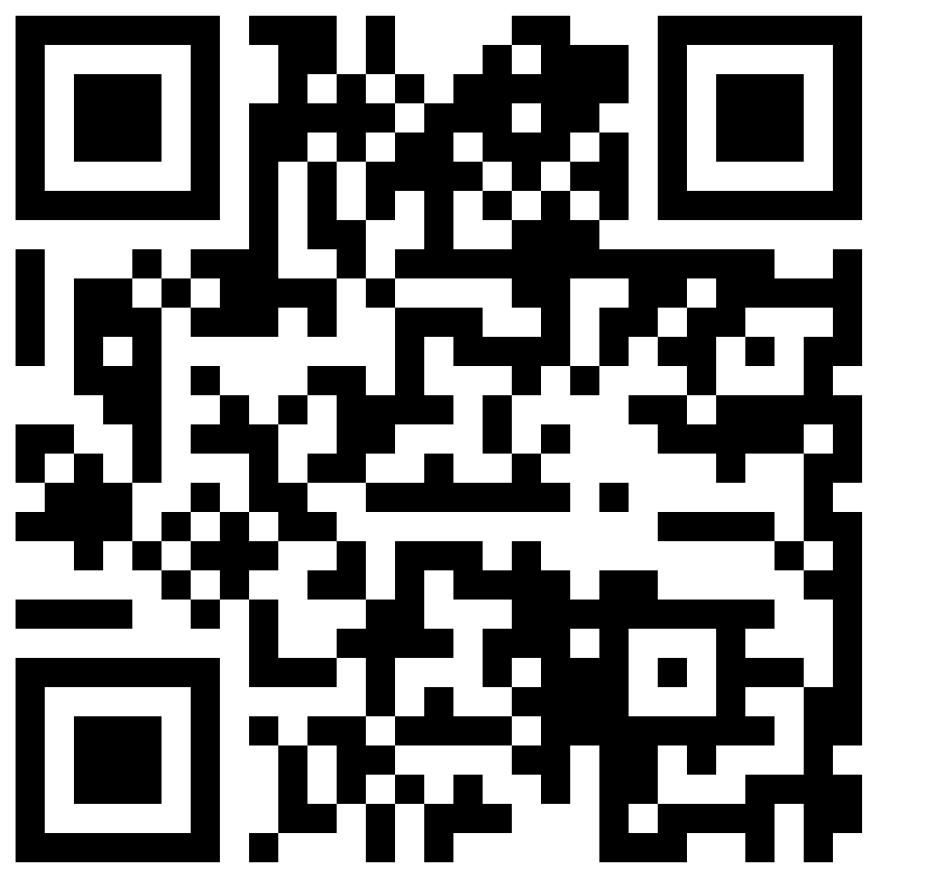


Humans revisit visual evidence. Multimodal LLMs should, too.

Project page

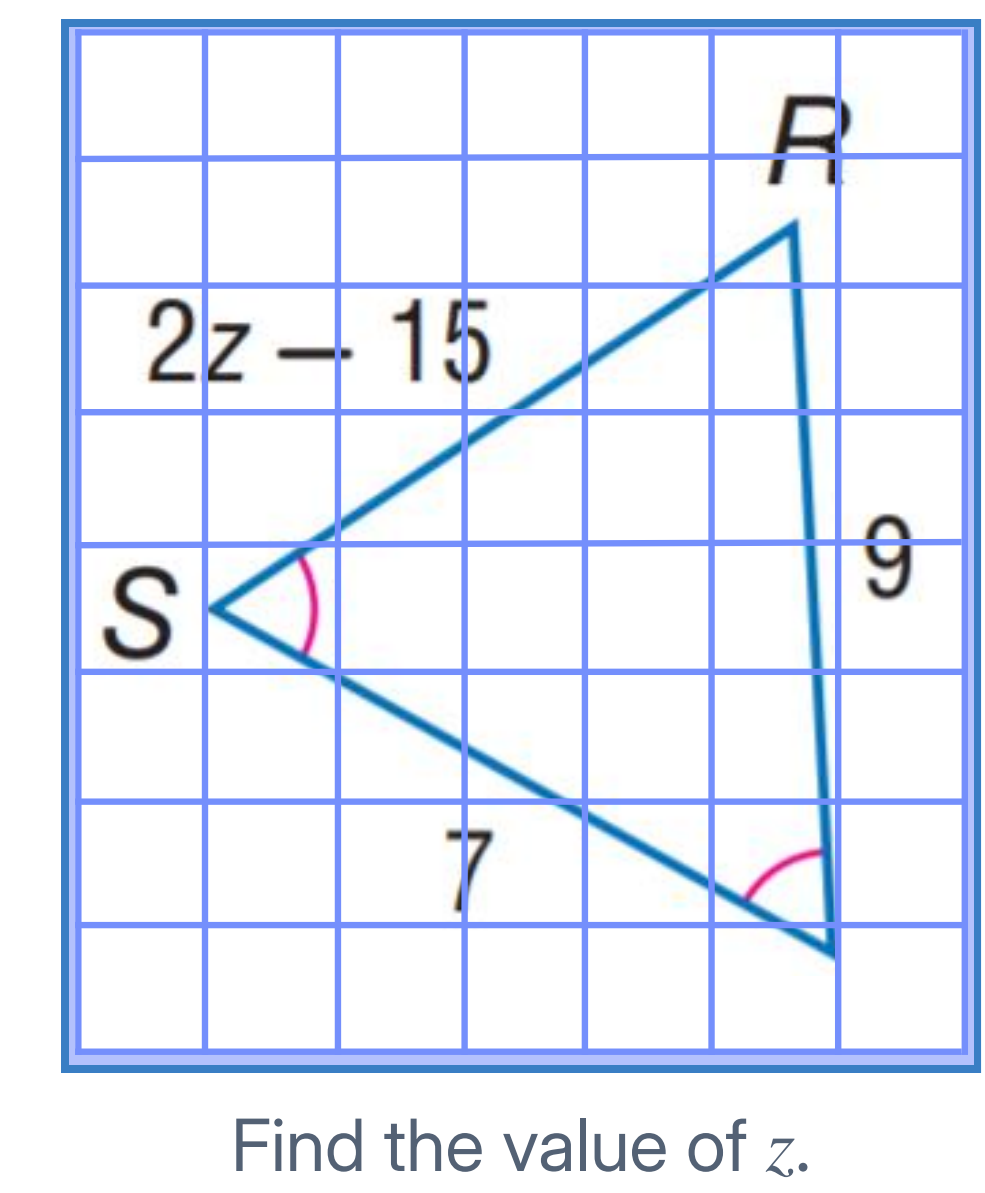


v1: Learning to Point Visual Tokens for Multimodal Grounded Reasoning

First authors
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MOTIVATION

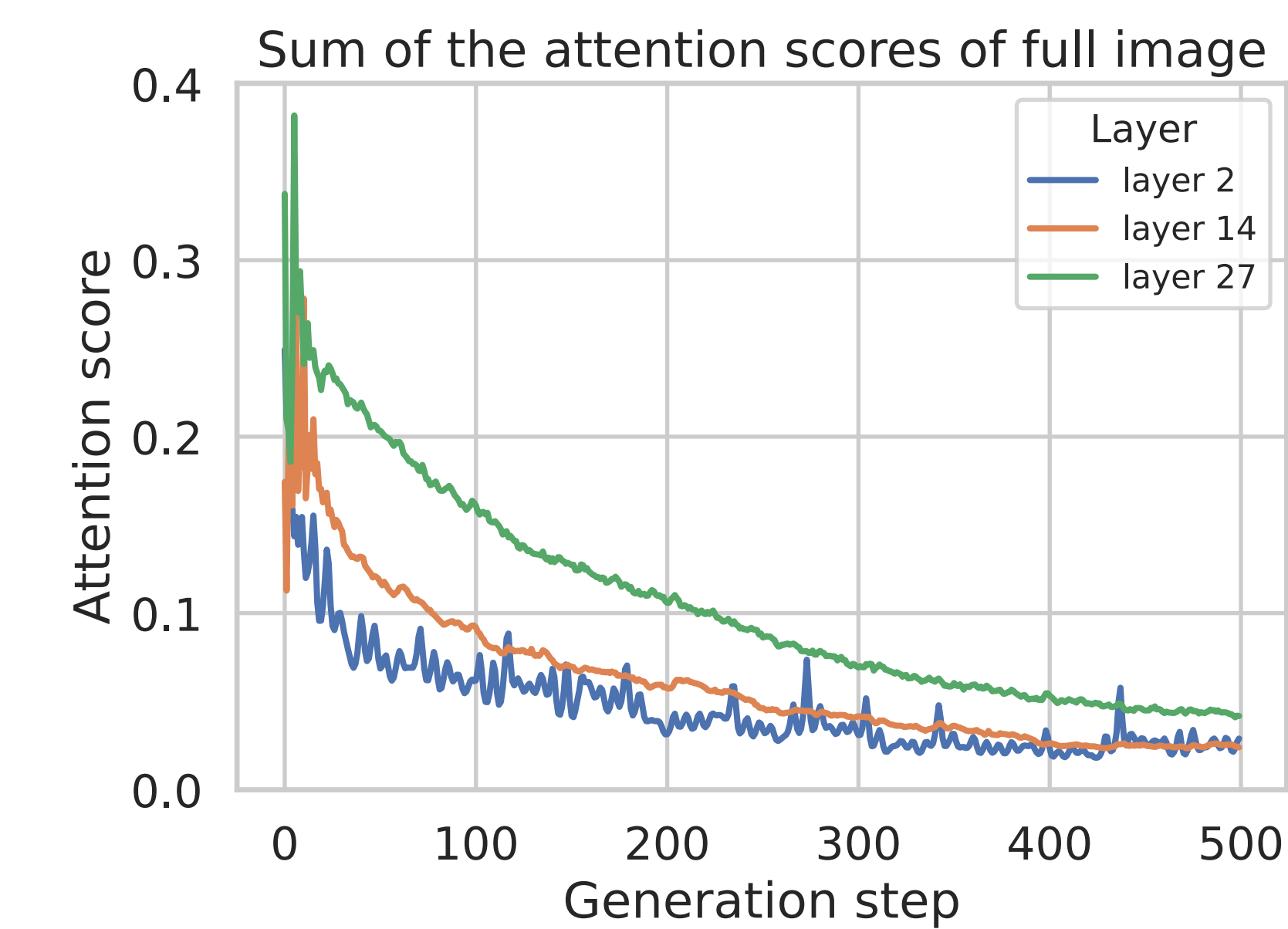
Visual attention fades during reasoning.
v1 explicitly revisits relevant patches.



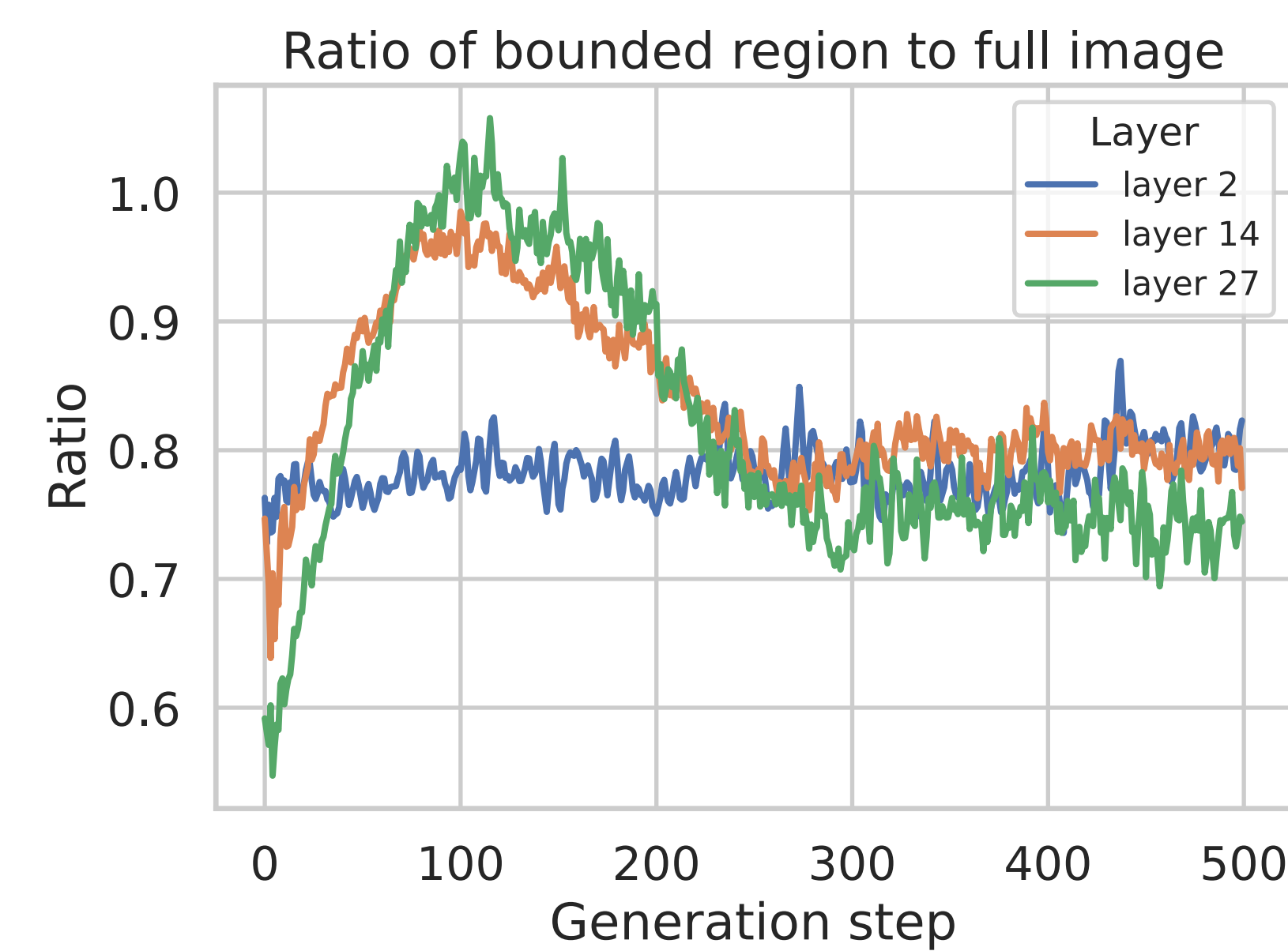
Text-based reasoning can attend to image tokens, but visual attention fades
"... In an isosceles right triangle, the two equal sides would be $2z - 15 = 7 \Rightarrow z = 11$." ✗

v1 points to and copies image regions throughout reasoning
"... detect(query="angle at S") <region> [S] [1] </region> the angles at S and T seem to be the same, so $RS = RT$. Thus $2z - 15 = 9 \Rightarrow z = 12$." ✓

TVC-7B on RefCOCO testA · attention from the newest token to the image tokens · layers 2, 14, 27 · paper Fig. 3



Total attention on image tokens per step.
Declines steadily throughout decoding.

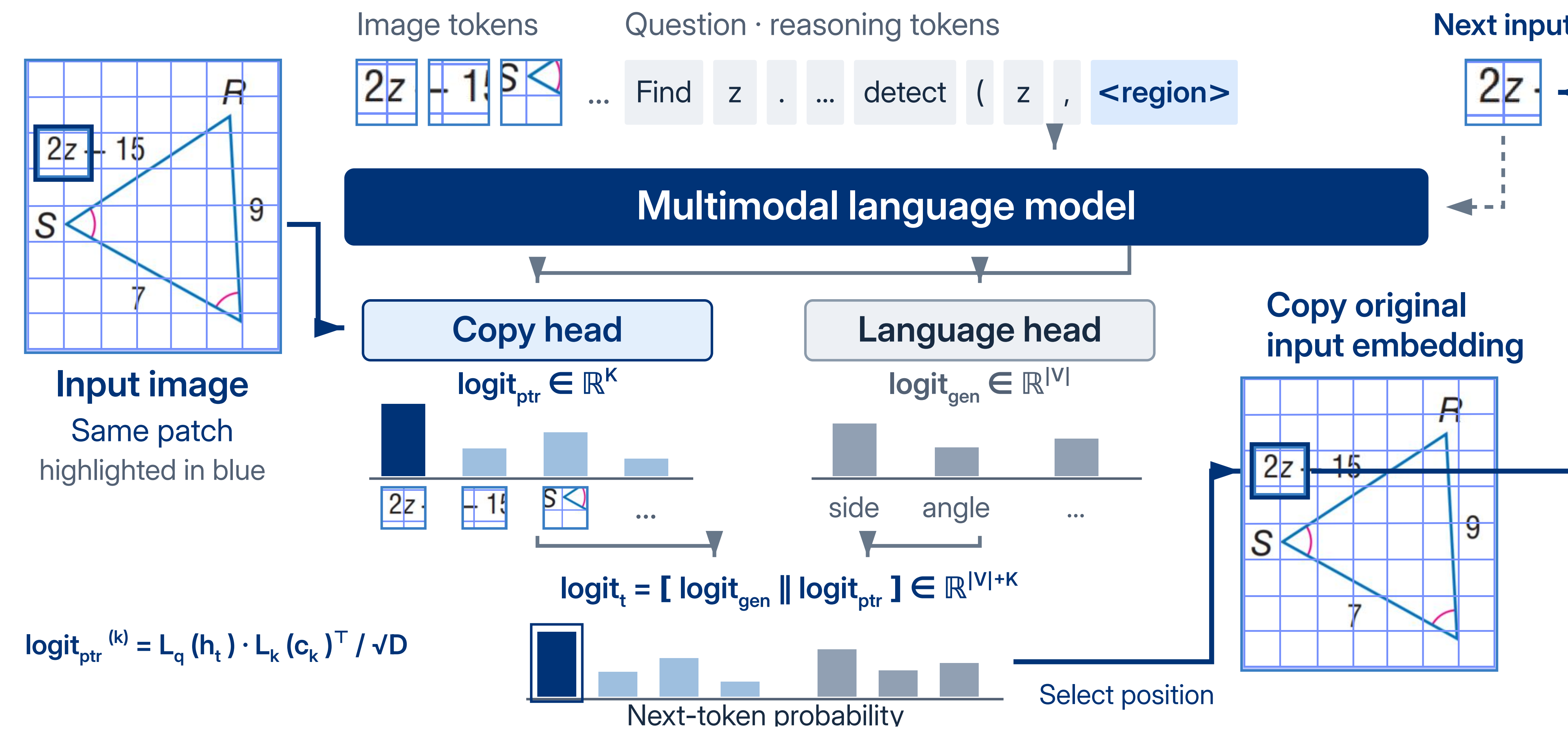


Relative target-region attention: target-box mean / all-image-token mean.

Converges to ≈ 0.8 by mid-generation, below average.

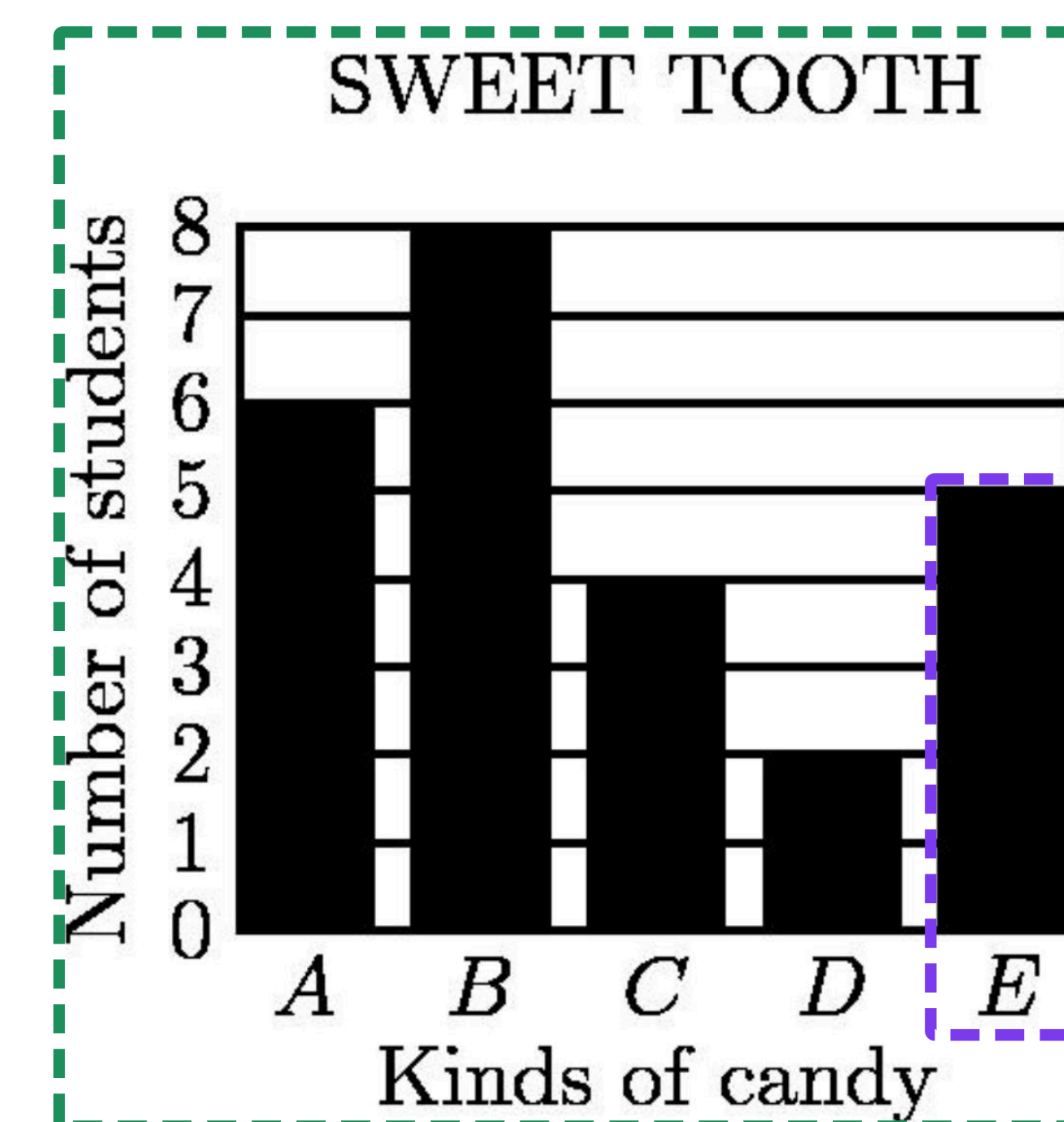
METHOD · POINT-AND-COPY

Generate a word, or point to a patch and copy it.



- Pointing logit $\text{logit}_{\text{ptr}}^{(k)}$: copy-head score for patch k , a query from the hidden state h_t against a key from the patch embedding c_k ; L_q, L_k are the only new parameters.
- Training · v1g. 300K reasoning traces with interleaved grounding annotations, on Qwen2.5-VL-7B.

QUALITATIVE EXAMPLE



Let's see here. I've got this bar graph.
detect(query="bar graph titled \"Sweet Tooth\"", objects=["<obj1> <region> [0.23, 0.25, 0.96, 0.79] <|copy_76|> ... <|copy_287|> </region>"])

The bar graph <obj1> shows the number of students labeled from A to E.
... (y-axis look-up omitted) ...

detect(query="bar for Candy E", objects=["<obj8> <region> [0.74, 0.37, 0.96, 0.88] <|copy_139|> ... <|copy_303|> </region>"])

Candy E <obj8> has a bar reaching up to 5.
There are 25 students in total. So, percentage = $(5 / 25) * 100 = 20\%$

RESULTS · MULTIMODAL MATH REASONING

Outperforms 7B baselines; narrows the gap to 72B models.

Model	Size	Reasoning Only	MathVista mini	MathVision full	MathVerse mini	Average mini	Average full
Qwen2-VL (Wang et al., 2024b)	7B	✗	60.9	-	16.3	24.6	-
Qwen2-VL (Wang et al., 2024b)	72B	✗	69.7	-	26.6	36.2	-
Qwen2.5-VL (Bai et al., 2025)	7B	✗	67.8	23.6	-	44.5	45.3
Qwen2.5-VL (Bai et al., 2025)	72B	✗	74.8	39.8	-	57.6	57.4
InternVL2.5 (Chen et al., 2025)	8B	✗	64.4	22.0	19.7	39.5	41.9
InternVL2.5 (Chen et al., 2025)	78B	✗	72.3	34.9	32.2	51.7	53.0
GPT-4o (Hurst et al., 2024)	-	✗	63.8	-	30.4	50.2	-
LLaVa-CoT (Xu et al., 2025)	11B	✓	54.8	16.3	-	33.9	35.0
Mulberry (Yao et al., 2024)	7B	✓	63.1	-	-	39.6	-
TVC (Sun et al., 2025)	7B	✓	68.1	-	22.7	38.9	-
TVC (Sun et al., 2025)	72B	✓	72.2	-	41.9	48.8	-
QVQ-72B-preview (Qwen Team, 2024)	72B	✓	71.4	35.9	-	41.5	-
Base (Qwen2.5-VL)	7B	✗	67.8	23.6	-	44.5	45.3
Text-Only (TVC)	7B	✓	68.1	-	22.7	38.9	-
Ours	7B	✓	68.6	34.5	28.1	48.6	50.6
↳ Inference w/o Pointing	7B	✓	60.0	25.3	23.7	33.6	39.6

Table 1 · accuracy (%), Gemini-2.0-Flash judge · bold: best 7B-scale result · 72B/78B and GPT-4o are reference points

ANALYSIS

Variant	Tr.	Inf.	Score	Δ
Backbone	✗	✗	23.6	-
(no pointing)	✓	✗	25.3	+1.7
Ours	✓	✓	34.5	+10.9

Pointing at inference drives the gain.

Disabling point-and-copy at inference drops MathVision-mini accuracy from 34.5 to 25.3.

v1 sees the image again: attention lands on the copied patches.

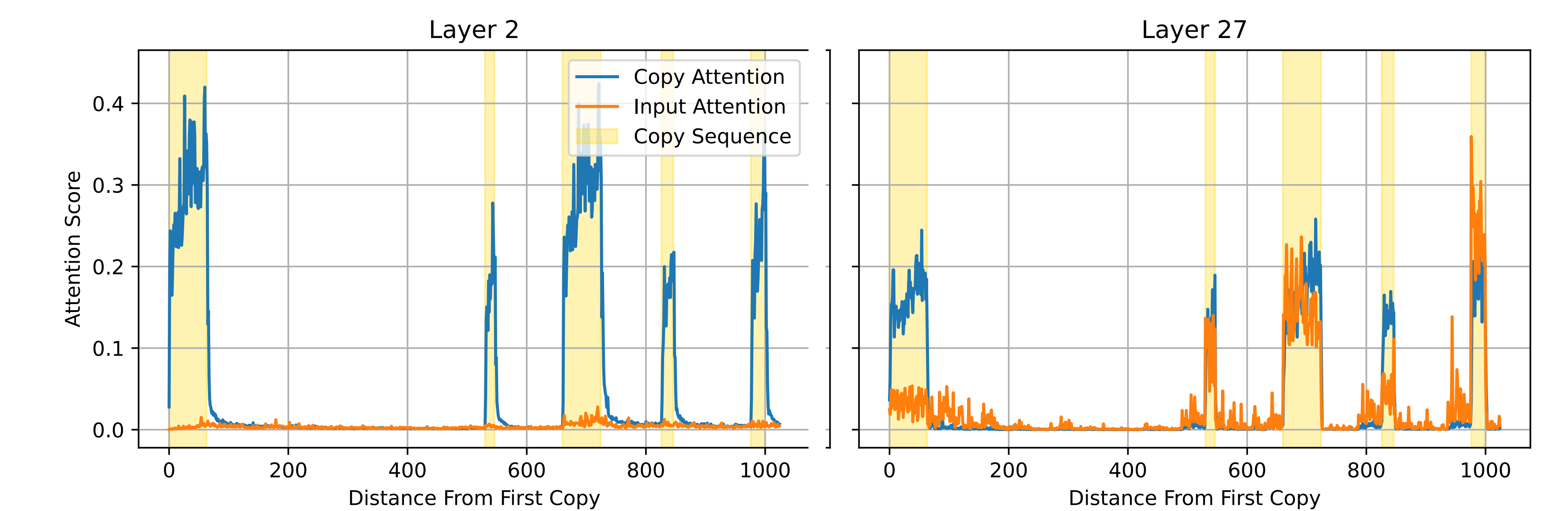


Fig. 5 · attention to copied vs. original visual tokens after the first copy. Layers 2 and 14 attend mostly to the copied tokens; layer 27 balances copied and original tokens.